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To cite this article: G N De Side *et al* 2025 *IOP Conf. Ser.: Earth Environ. Sci.* **1441** 012012

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IoT (internet of things) microclimate monitoring system using node-red platform at plant factory

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Abstract. Indonesia faces significant challenges in the agricultural sector, such as limited productive land and climate change, which threaten food security. The plant factory emerges as a relevant solution by integrating hydroponic technology and automation to enable year-round controlled food production. This study aimed to develop an IoT-based microclimate monitoring system using the Node-RED platform within a plant factory. The system allowed for real-time temperature, humidity, and light intensity monitoring. The methodology involves designing a system utilizing an Arduino Mega 2560 microcontroller, DHT22 and GY-49 sensors, and an ESP01S WiFi module. The collected data was transmitted via the MQTT protocol and displayed on the Node-RED dashboard. The results demonstrated that this system effectively monitored climate parameters and could enhance the efficiency and productivity of the plant factory, represented by all sensors functioning correctly. The DHT22 sensor has been calibrated with a standard Thermohygrometer measuring instrument with a temperature RMSE value of 1.01 and a humidity RMSE of 5.88. Likewise, the GY-49 sensor has been calibrated with a standard lux meter. In calibrating the GY-49, the equation entered in the coding is $0.978 * \text{sensor reading} + 0.891$, and the resulting RMSE value is 57.91. During the testing phase, the datalogger hardware system can run 100%. Consequently, this system significantly supports sustainable modern agriculture and national food security.

1. Introduction

As an agricultural country with a large population, Indonesia faces significant challenges in the agricultural sector that drive the urgency of developing modern crop cultivation technologies such as plant factories [1]. One of the main problems faced is the limitation of productive agricultural land. Rapid population growth and high urbanization rates have converted massive agricultural land into non-agricultural areas. Data from the Central Bureau of Statistics shows that around 100,000 hectares of agricultural land is converted yearly into industrial areas, housing, and other infrastructure [2]. This phenomenon threatens national food availability and potentially increases Indonesia's dependence on imports of certain food commodities [3] [4].

Global climate change is the next challenge that further complicates conventional agricultural practices in Indonesia [5]. Extreme weather phenomena such as prolonged droughts or unexpected floods have become more frequent, causing uncertainty in agricultural production [6]. Changing and unpredictable rainfall patterns directly impact planting and harvesting schedules and the quality of agricultural produce. The Ministry of Agriculture of the Republic of



Indonesia, in its Master Strategy for Agricultural Development 2015-2045, recognizes that climate change poses a severe threat to national food security and requires a comprehensive adaptation strategy [7].

In addition, conventional agricultural practices in Indonesia still rely heavily on the overuse of pesticides and other chemicals [8]. Although these chemical inputs aim to increase crop yields and control pests, their long-term impact on human health and the environment cannot be ignored. Soil and water pollution due to pesticide residues has become a severe problem in many agricultural areas in Indonesia [9]. This not only threatens biodiversity but also potentially jeopardizes consumer health through the accumulation of chemical residues in agricultural products.

Water use efficiency in conventional agriculture is also of particular concern. Indonesia, although known as a country with high rainfall, faces the problem of uneven water distribution between regions and seasons. Some areas experience water scarcity, especially during the dry season, while the agricultural sector is the largest consumer of water. Inefficient irrigation practices in conventional agriculture further exacerbate this situation, threatening future food production sustainability [10][11]. In this context, the development of plant factories has become highly relevant and essential for Indonesia. A plant factory is a fully controlled indoor plant cultivation system that uses artificial lighting, hydroponics and automation technologies to optimize plant growth. The concept offers several advantages that can overcome the challenges of conventional farming in Indonesia.

Plant factories facilitate year-round food production, independent of weather conditions or seasonal variations. This capability is especially crucial in the context of increasing climate uncertainty [12] [13][14]. The plant factory system can be built vertically to save land. In addition, regulating water delivery will provide a higher efficiency value, and pest attacks can be minimized because they use a closed system. Plant factories require a precise environmental control system to ensure production stability and consistent crop quality, providing the certainty of food supply needed to support national food security.

The development of plant factories is currently leading to IoT (Internet of Things) based remote monitoring systems [15]. IoT systems enable precise monitoring and control of various aspects of the plant growth environment, from temperature, humidity, and light intensity to CO₂ and nutrient levels. Environmental data can be collected in real-time through a network of connected sensors, giving operators an in-depth insight into plant growth conditions. As technology develops and the cost of IoT devices decreases, adoption of this technology in plant factories is expected to continue to increase. IoT integration not only improves efficiency and productivity but also enables a level of precision and control that is difficult to achieve in conventional agriculture [16], paving the way for food production that is more sustainable and responsive to dynamic market needs.

These references show that there have been many previous studies on plant factories and IoT, but in this research, the concept is not only developing a plant factory system but also supported by IoT using Node-RED, which has the advantage of using a visual programming language that focuses on the program as a flow. Considering the complexity of agricultural challenges faced by Indonesia and the potential solutions offered by plant factories, this research will develop an IoT-based microclimate monitoring system using Node-RED to strengthen national food security, promote modern agriculture, and support sustainable development (SDGs).

2. Method

2.1 Equipment and materials

The plants used in this research were grand rapid lettuce seeds (*Lactuca sativa* L.), AB mix solution specifically for hydroponic vegetable plants, and rock wool. As for the design of the IoT monitoring system, it requires Arduino Mega 2560 microcontroller hardware, datalogger shield, 20x4 LCD with I2C, DHT-22 sensor, GY-49 sensor, ESP8266 ESP01S Serial Wireless Module WiFi sensor, ESP8266 ESP-01 Serial WiFi wireless Adapter Module, 9 V/1 A adapter. Software to run the program using Arduino IDE version 2.3.2.

2.2 Monitoring system design

The climate monitoring system in the plant factory was designed with several main components. As input, the DHT22 sensor was used with the VCC pin connection to the Arduino Mega 5 Volt pin, and the ground pin was connected to the Arduino Mega Ground pin, the data pin was connected to pin 2 of the Arduino Mega. The GY-49 sensor has four pins. Each VCC pin was connected to the Arduino Mega 5 Volt pin, and the ground pin was connected to the Arduino Mega Ground pin, the SCL pin to the Arduino Mega SCL pin, the SDA pin to the Arduino Mega SDA pin. ESP01S served as a WiFi connection consisting of 4 pins. The VCC pin was connected to the 3 Volt Arduino Mega pin. The ground pin was connected to the Arduino Mega ground pin. The RX pin was connected to the Arduino Mega TX pin, and the TX pin was connected to the Arduino Mega RX pin. The 20x4 LCD was connected with four pins, each consisting of a VCC pin connected to a 5 Volt Arduino Mega pin, a ground pin connected to a ground Arduino Mega pin, an SDA pin to pin four Arduino Mega, and an SCL pin connected to pin 5 Arduino mega. The installation position can be seen in Figure 1.

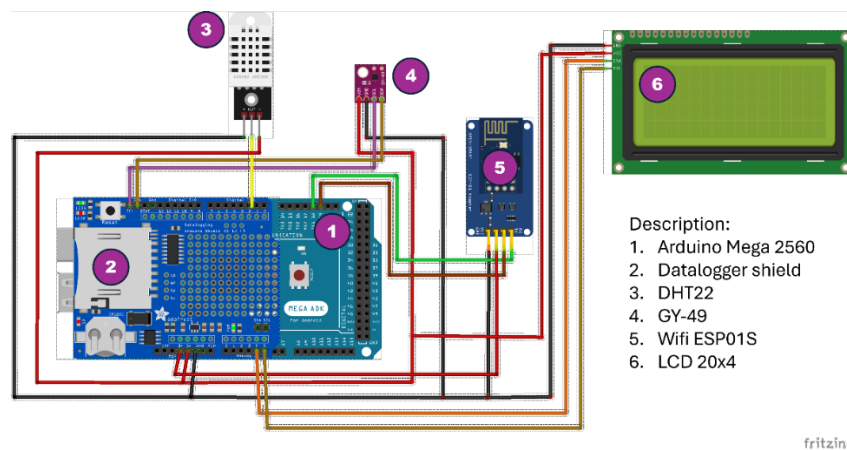


Figure 1. Climate monitoring system design

Figure 1 explains the data logger shield connected to the Arduino Mega 2560 with some modifications because the data logger shield used was only designed for Arduino Uno, so modifications must be made first. Setting (data storage) digital pins 11, 12, and 13 on the Data logger shield were not connected directly, but the legs were cut or bent. Furthermore, pin 11 was connected to digital pin 51 (MOSI), pin 12 to pin 50 (MISO), and pin 13 was connected to 52 (SCK). Meanwhile, pin 10 remains connected to pin 10 Arduino Mega. A4 (SDA) and A5 (SCL) were connected to pins 20 and 21. VCC and GND were connected to 5V and GND pins.

2.3 MQTT and node-red communication protocol

MQTT, or Message Queuing Telemetry Transport, was a protocol with client-server publish/subscribe properties. MQTT was a transport protocol with simple, open, and lightweight characteristics that were designed to be easy to implement [17]. So, MQTT can be used in many situations, including machine-to-machine (M2M) and Internet of Things (IoT) communications. This publish-subscribe messaging pattern required a message broker. The message broker was responsible for distributing messages to clients based on the topics that each sensor outputs. This research has 3 topics: keywords for connecting sensors to unordered via MQTT Broker. The 3 topics are 'node/mega/dht22/temperature', 'node/mega/dht22/temperature', and 'node/mega/dht22/temperature'. The MQTT broker reads these 3 topics and then sent further to the Node-Red system.

For data display and recording, this research uses Node-RED. Node-RED is software for connecting hardware based on a web-based editor. Node-RED is programmed in the form of flows with various pallet features. Figure 2 shows one of the flows for reading the temperature sensor in the greenhouse. The flow starts with the MQTT block, which reads the sensor data from the corresponding topic. This process is called the subscribe process.

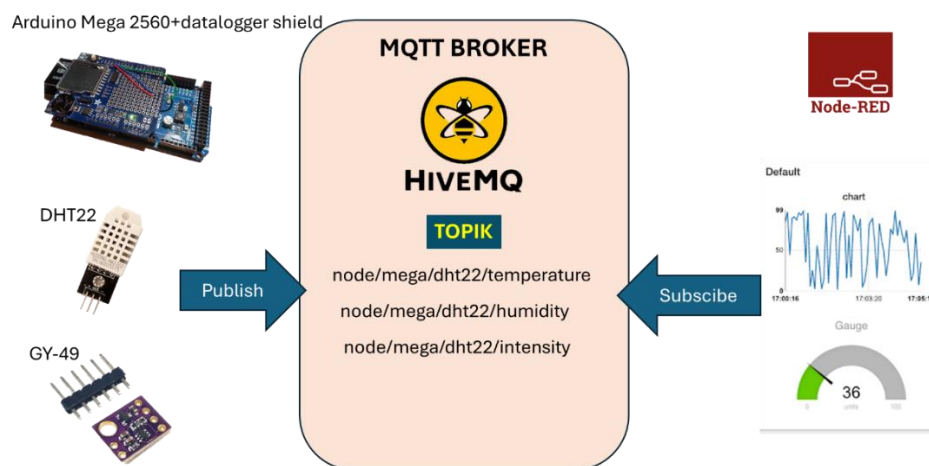


Figure 2. MQTT real protocol

The microclimate detection module was made from 1 Arduino Mega 2560 microcontroller with 2 DHT22 and GY-49 sensors connected. The ESP01S WiFi router connects communication between the Arduino Mega 2560 microcontroller and activates the Node-RED dashboard, accessed via a PC/laptop. The read data sent by the sensor is then processed by ESP01S and forwarded to the MQTT broker. If the detection module was turned on with an adapter voltage of 9 Volts and a current of 1 A, the sensor will read the state of the room. The sensor sent the read data, processed by ESP01S, and forwarded by the MQTT Broker, which has been connected to the Node-Red application.

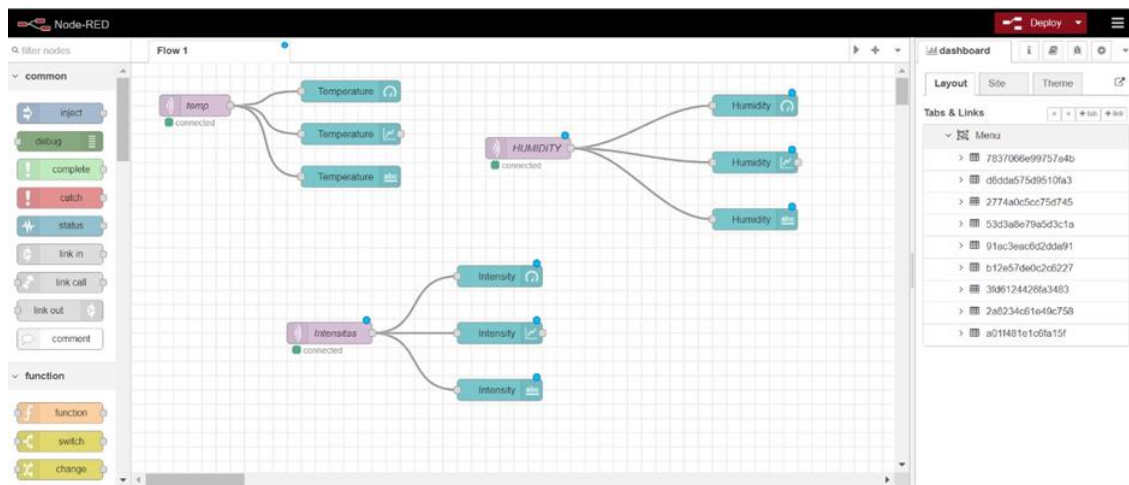


Figure 3. Flow in the Node-RED development environment

Node-RED was a browser-based tool for creating Internet of Things (IoT) applications whose visual programming environment makes it easy for users to create applications as a “flow” [18]. Node-RED took an alternative path in software development. The advantage of Node-Red was that it uses a visual programming language. Instead of creating applications as lines of code, Node-RED focused on the program as a flow. This flow comprised interconnected nodes, each performing a specific task (Figure 3). Although Node-RED was designed for the Internet of Things (IoT), it can also be used for general purposes and various applications.

2.4 System workflow

The Arduino Mega 2560 device was the monitoring centre of the plant factory. This device was connected to the datalogger, LCD monitor, ESP01S, and sensors. The source of electrical energy used to run this system uses a supply from an adapter, while the command and information transmission process uses a local host server connected to the Node-Red application device with the help of the MQTT protocol, as shown in Figure 4.

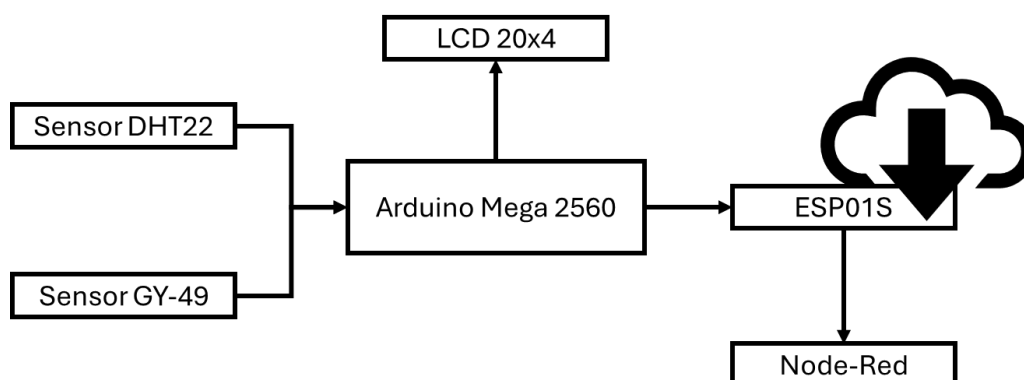


Figure 4. System flow design

The process started with sensor readings by the ESP01S. The 3 sensor readings were given different topic names. The readings were taken within 120 seconds. The sensor readings were sent to Node-RED and installed on the Windows system using the MQTT communication protocol. Node-RED then displayed the data on the dashboard.

3. Results and Discussion

Testing is carried out after the monitoring system has been assembled and configured on each component. The tests carried out consisted of 4 stages, namely, 1) testing sensor readings, 2) sending sensor data via WiFi3) data storage, and 4) testing in the plant factory.

3.1. Sensor reading results

Sensor reading tests are performed before installing the datalogger shield and ESP01S modules. This test aims to ensure that all sensors are functioning correctly. Testing is done by connecting the sensor to the Arduino Mega 2560, then through the Arduino IDE, the sensor reading program is uploaded, and the sensor readings are displayed via a serial monitor. The sensor reading test is shown in Table 1.

Table 1. Sensor reading results

Date	Time	Temperature (°C)	Humidity (%)	Intensity (Lux)
7/2/2024	10:50:56	30.2	76.6	2718.72
7/2/2024	10:53:01	30.2	77.4	2730.24
7/2/2024	10:55:07	30.1	76.9	2730.24
7/2/2024	10:57:12	30.1	78	2718.72
7/2/2024	10:59:18	30.2	76.8	2718.72
7/2/2024	11:01:23	30.2	78.8	2718.72
7/2/2024	11:03:29	30.2	77.3	2730.24
7/2/2024	11:05:35	30.2	78.1	2718.72

Table 1 shows that all sensors function correctly, as seen from the values displayed on the serial monitor. The DHT22 sensor takes temperature and humidity readings well, and the GY-49 sensor can take light intensity readings well. The DHT22 sensor has been calibrated with a standard Thermohigrometer measuring instrument with a temperature RMSE value of 1.01 and a humidity RMSE of 5.88. Likewise, the GY-49 sensor has been calibrated with a standard lux meter. In calibrating the GY-49, the equation entered in the coding is $0.978 * \text{sensor reading} + 0.891$, and the resulting RMSE value is 57.91.

3.2. Sensor storage to datalogger shield

The datalogger design is presented in Figure 5. Datalogger testing begins by ensuring that the designed data logger can record data read by the sensor. The data logger presents real-time time, temperature, humidity, and light intensity data. This data logger test is intended to get real-time data with reasonable accuracy when implemented in the plant factory.

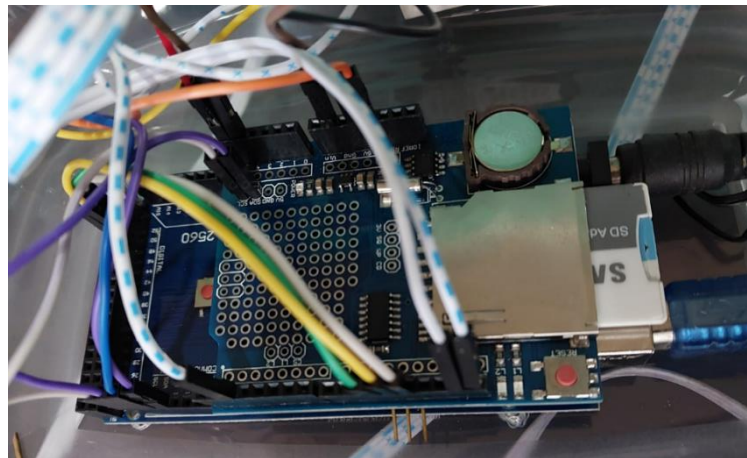


Figure 5. Configuration of datalogger shield with Arduino Mega 2560

During the testing phase, the datalogger hardware system can run 100%, characterized by data that can be written to the SD card with the .csv file format, and the RTC can present the time when reading the sensor measurement data. Data storage is done in real time, with data storage time set every 2 minutes. The data stored on the SD Card file is set in date, day, and time format then accompanied by the sensor measurement results. In this study, two sensors were utilized to collect measurements of temperature ($^{\circ}\text{C}$), humidity (%), and light intensity (lux).

3.3. Sensor data transmission to gateway

After the sensor has successfully taken readings, test the sending of sensor data from the device (Arduino Mega 2560 Module) to the gateway via ESP01S. The program code inputted through the Arduino IDE is used to read parameters or data on the sensors that have been determined and then send the data via the internet to the MQTT server. On the MQTT server side, the Things Board IoT platform is used, which is helpful to an MQTT broker and data presenter in the form of a Node-Red dashboard. The data displayed on the Node-Red dashboard can be in line or gauge charts to facilitate the monitoring process. The test results of sending sensor data from the device to the gateway are displayed through the serial monitor of the Arduino IDE application, as shown in Figure 6.

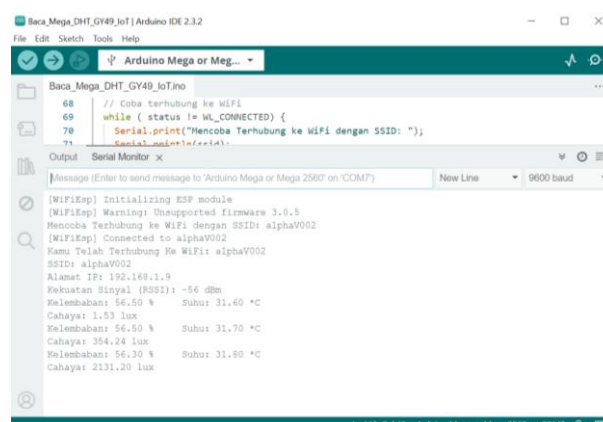


Figure 6. Sensor readings via gateway connection in Arduino IDE

The information obtained from the serial monitor in Figure 6 is that the device connection to the gateway is successful, based on the Connected to MQTT message that can be seen in the MQTTBox application. In the MQTTBox application, add subscribers and fill in the topics inputted during the WiFi connection for the MQTT client using the server from MQTT broker.hivemq.com. Each topic displays sensor data, and the topics cannot be the same.

3.4 Testing in the plant factory

The next test is to install the IoT device in the plant factory. The IoT device is placed near the plant factory and connected to PLN electricity using a 9 Volt, 1 A adapter.



Figure 7. Installation of climate monitoring system in the plant factory

The device, as shown in Figure 7 is built in one box. The components in the box include a GY-49 sensor to monitor light intensity and a DHT22 sensor to monitor temperature and humidity in the plant factory. The ESP01S WiFi module must be installed in an area with a strong enough WiFi signal to avoid lost connections while monitoring.

3.5 Dashboard on Web Application

Environmental data is read by the sensor every 2 minutes. The data readings are then sent to the gateway via a wireless network. Then, the data is displayed with an interface as a web application using the gateway's Node-RED Dashboard. The interface of the system for monitoring environmental conditions on farmland is shown in Figure 8.

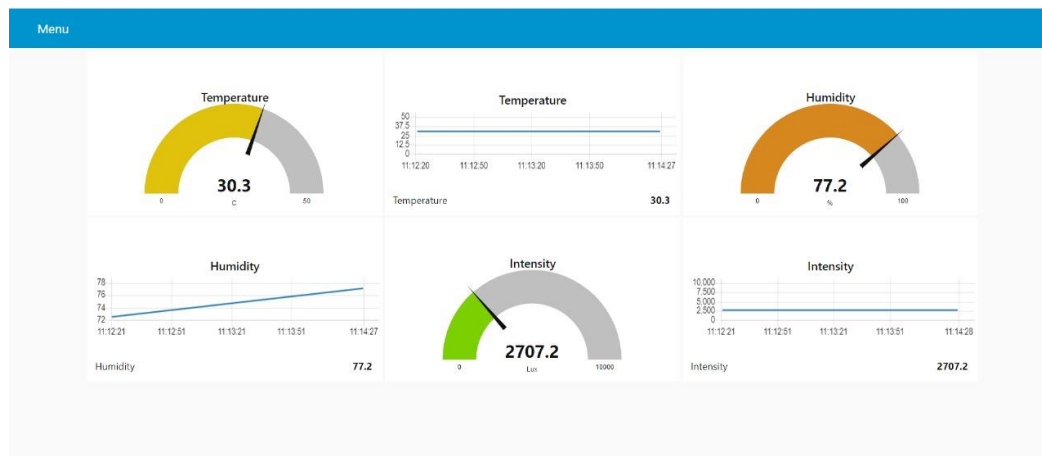


Figure 8. Dashboard view of the monitoring system through Node-Red

From the dashboard display on Node Red, it can be seen that the temperature in the plant factory is an average of 29.068 °C. The highest temperature is found in the afternoon around 2:00 p.m., which is 31.4 °C, and the lowest is 26.5 °C at 04:30 a.m. The humidity graph shows that the plant factory has an average humidity of 77.52%, with the highest humidity of 83.6% at 2:40 pm and the lowest humidity at 3:27 pm, 66.2%. The light intensity parameter shows that the average light intensity in the plant factory is 2158.05 lux. There is no significant change in the light intensity because the light source comes from the grow light with a stable light intensity level. The lights are off from 00.30 to 4.30 pm, so the intensity value is zero.

4. Conclusions

In this study, a sensor node network monitoring system based on the MQTT protocol has been successfully created. The designed sensor node can measure temperature, humidity, and solar intensity parameters. The monitoring system on the IoT-based plant factory with Node-RED has been designed. It can display all sensor data sent via the Node-RED dashboard with high efficiency and better productivity compared to conventional farming methods. Its impact on sustainable agriculture in Indonesia is very positive, especially in increasing crop yields and adaptation strategies to uncertain climate change. However, the implementation of this system requires support in terms of infrastructure, technical training, and government policies that support high-tech agriculture.

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